**1** (a) Factorise $a^2 + 3a - 28$

28

1 28

2 14

4 7

$(a+7)(a-4)$

**(2)**

(b) Solve $a^2 + 3a - 28 = 0$

$a = -7$ or $a = 4$

**(1)**

**(Total for Question 1 is 3 marks)**

---

**2** (a) Factorise $x^2 - 7x + 10$

10

1 10

2 5

$(x-2)(x-5)$

**(2)**

(b) Solve $x^2 - 7x + 10 = 0$

$x = 2$ or $x = 5$

**(1)**

**(Total for Question 2 is 3 marks)**

---

**3** (a) Factorise $b^2 + 9b + 20$

20

1 20

2 10

4 5

$(b + 4)(b + 5)$

**(2)**

(b) Solve $b^2 + 9b + 20 = 0$

$b = -4$ or $b = -5$

**(1)**

**(Total for Question 3 is 3 marks)**

---

**4** (a) Factorise $x^2 - 3x - 18$

18

1 18

2 9

3 6

$(x + 3)(x - 6)$

**(2)**

(b) Solve $x^2 - 3x - 18 = 0$

$x = -3$ or $x = 6$

**(1)**

**(Total for Question 4 is 3 marks)**

**5** (a) Factorise $y^2 - 10y + 9$

9

1 9

3 3

$(y-1)(y-9)$

**(2)**

(b) Solve $y^2 - 10y + 9 = 0$

$y = 1$ or $y = 9$

**(1)**

**(Total for Question 5 is 3 marks)**

---

**6** (a) Factorise $a^2 - a - 56$

56

1 56

2 28

4 14

7 8

$(a+7)(a-8)$

**(2)**

(b) Solve $a^2 - a - 56 = 0$

$a = -7$ or $a = 8$

**(1)**

**(Total for Question 6 is 3 marks)**

**7** Solve $x^2 + 14x + 24 = 0$

| | |
|---|---|
| | 24 |
| 1 | 24 |
| 2 | 12 |
| 3 | 8 |
| 4 | 6 |

$(x + 2)(x + 12) = 0$

$x = -2 \quad x = -12$

$x = -2$ or $x = -12$

**(Total for Question 7 is 3 marks)**

---

**8** Solve $x^2 + 5x - 6 = 0$

| | |
|---|---|
| | 6 |
| 1 | 6 |
| 2 | 3 |

$(x - 1)(x + 6) = 0$

$x = 1 \quad x = -6$

$x = 1$ or $x = -6$

**(Total for Question 8 is 3 marks)**

---

**9** Solve $x^2 + 5x + 6 = 0$

$(x + 2)(x + 3) = 0$

$x = -2 \quad x = -3$

$x = -2$ or $x = -3$

**(Total for Question 9 is 3 marks)**

**10** Solve $x^2 - 12x + 32 = 0$

32

| | |
|---|---|
| 1 | 32 |
| 2 | 16 |
| 4 | 8 |

$(x - 4)(x - 8) = 0$

$x = 4 \quad x = 8$

$x = 4$ or $x = 8$

**(Total for Question 10 is 3 marks)**

---

**11** Solve $x^2 + 19x + 90 = 0$

90

| | |
|---|---|
| 1 | 90 |
| 2 | 45 |
| 3 | 30 |
| 5 | 18 |
| 6 | 15 |
| 9 | 10 |

$(x + 9)(x + 10) = 0$

$x = -9 \quad x = -10$

$x = -9$ or $x = -10$

**(Total for Question 11 is 3 marks)**

---

**12** Solve $x^2 + 11x - 42 = 0$

42

| | |
|---|---|
| 1 | 42 |
| 2 | 21 |
| 3 | 14 |
| 6 | 7 |

$(x - 3)(x + 14) = 0$

$x = 3 \quad x = -14$

$x = 3$ or $x = -14$

**(Total for Question 12 is 3 marks)**

**13** Solve $a^2 - 10a + 16 = 0$

16

| | |
|---|---|
| 1 | 16 |
| 2 | 8 |
| 4 | 4 |

$(a - 2)(a - 8) = 0$

$a = 2 \quad a = 8$

$a = 2$ or $a = 8$

**(Total for Question 13 is 3 marks)**

---

**14** Solve $y^2 - 2y - 35 = 0$

35

| | |
|---|---|
| 1 | 35 |
| 5 | 7 |

$(y + 5)(y - 7) = 0$

$y = -5 \quad y = 7$

$y = -5$ or $y = 7$

**(Total for Question 14 is 3 marks)**

---

**15** Solve $x^2 + 3x - 54 = 0$

54

| | |
|---|---|
| 1 | 54 |
| 2 | 27 |
| 3 | 18 |
| 6 | 9 |

$(x + 9)(x - 6) = 0$

$x = -9 \quad x = 6$

$x = -9$ or $x = 6$

**(Total for Question 15 is 3 marks)**

**16** Solve $b^2 - 10b - 24 = 0$

| | |
|---|---|
| | 24 |
| 1 | 24 |
| 2 | 12 |
| 3 | 8 |
| 4 | 6 |

$(b+2)(b-12) = 0$

$b = -2 \quad b = 12$

$b = -2$ or $b = 12$

**(Total for Question 16 is 3 marks)**

---

**17** Solve $m^2 + 13m + 40 = 0$

| | |
|---|---|
| | 40 |
| 1 | 40 |
| 2 | 20 |
| 4 | 10 |
| 5 | 8 |

$(m+5)(m+8) = 0$

$m = -5 \quad m = -8$

$m = -5$ or $m = -8$

**(Total for Question 17 is 3 marks)**

---

**18** Solve $x^2 + 10x - 24 = 0$

| | |
|---|---|
| | 24 |
| 1 | 24 |
| 2 | 12 |
| 3 | 8 |
| 4 | 6 |

$(x+12)(x-2) = 0$

$x = -12 \quad x = 2$

$x = -12$ or $x = 2$

**(Total for Question 18 is 3 marks)**